EXP. NO: 05
DATE: 11/8/26

DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY.

Aim:

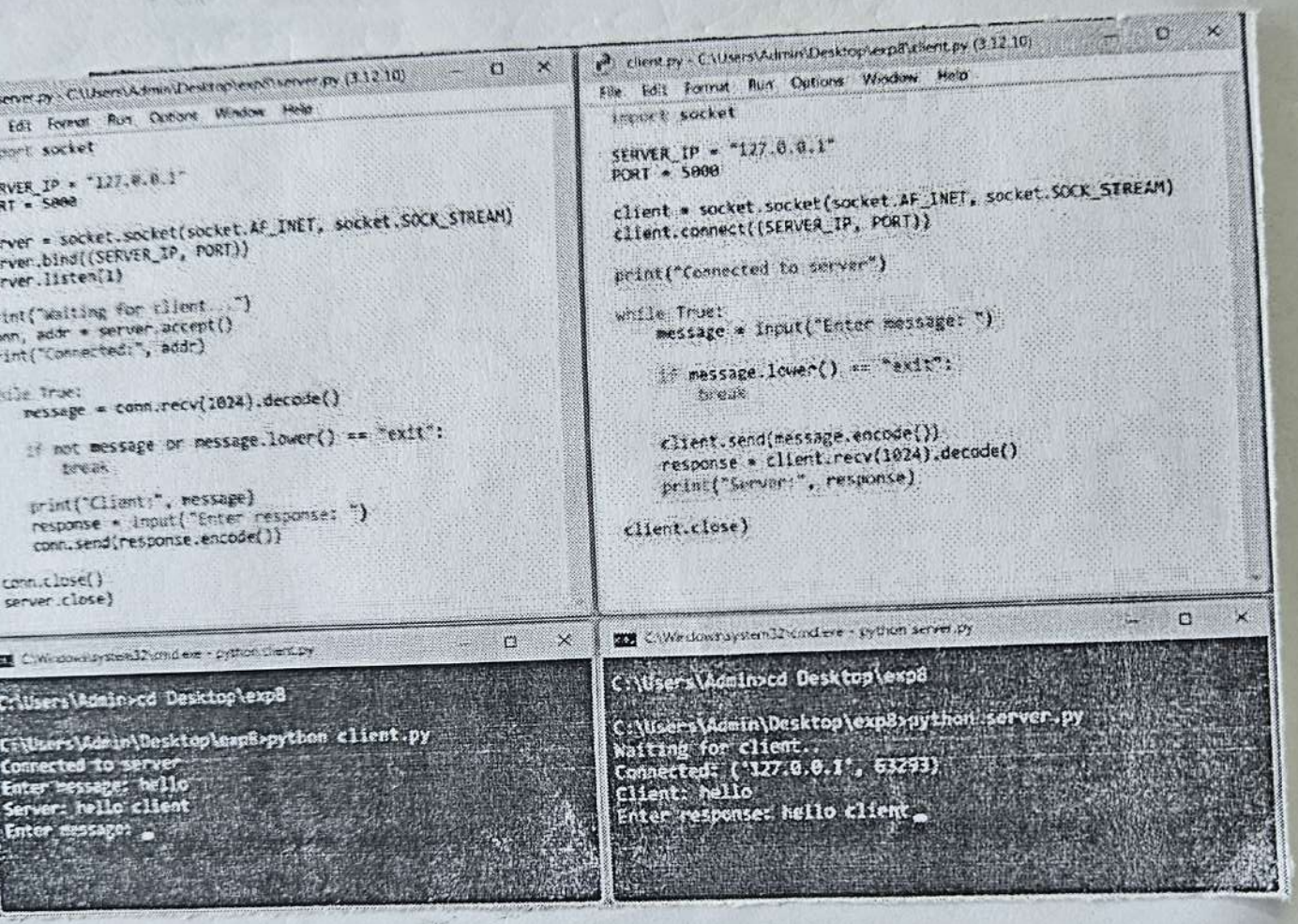
To develop a customized ping utility in python that checks connectivity b/w a client and a server by sending ICMP echo request packets and displaying response time, packet statistics and server reachability status.

Procedure:

1. Install python and Vs code.
2. Create ping.py file
3. Import subprocess and platform module
4. Accept hostname or IP address from user.
5. Detect OS using platform module.
6. Ping command:
   - Ping -n 4 <host> for windows
   - Ping -c 4 <host> for linux
7. Use the subprocess run() function to execute ping command capture output
8. Display ping results.
9. Display "server status" ; Reachable" if ping successful ; otherwise : "Server Status unreachable".

Result:

The customized ping program was executed successfully.



```
Edit Format Run Options Window Help
port socket

RVER_IP = "127.0.0.1"
RT = 5000

rver = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
rver.bind((SERVER_IP, PORT))
rver.listen(1)

int("Waiting for client...")
nn, addr = server.accept()
int("Connected:", addr)

hile True:
  message = conn.recv(1024).decode()

  if not message or message.lower() == "exit":
    break

  print("Client:", message)
  response = input("Enter response: ")
  conn.send(response.encode())

conn.close()
server.close()
```

```
File Edit Format Run Options Window Help
import socket

SERVER_IP = "127.0.0.1"
PORT = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((SERVER_IP, PORT))

print("Connected to server")

while True:
    message = input("Enter message: ")

    if message.lower() == "exit":
        break

    client.send(message.encode())
    response = client.recv(1024).decode()
    print("Server:", response)

client.close()
```

```
C:\Users\Admin>cd Desktop\exp8

C:\Users\Admin\Desktop\exp8>python client.py
Connected to server
Enter message: hello
Server: hello client
Enter message:
```

```
C:\Users\Admin>cd Desktop\exp8

C:\Users\Admin\Desktop\exp8>python server.py
Waiting for client...
Connected: ('127.0.0.1', 63293)
Client: hello
Enter response: hello client
```